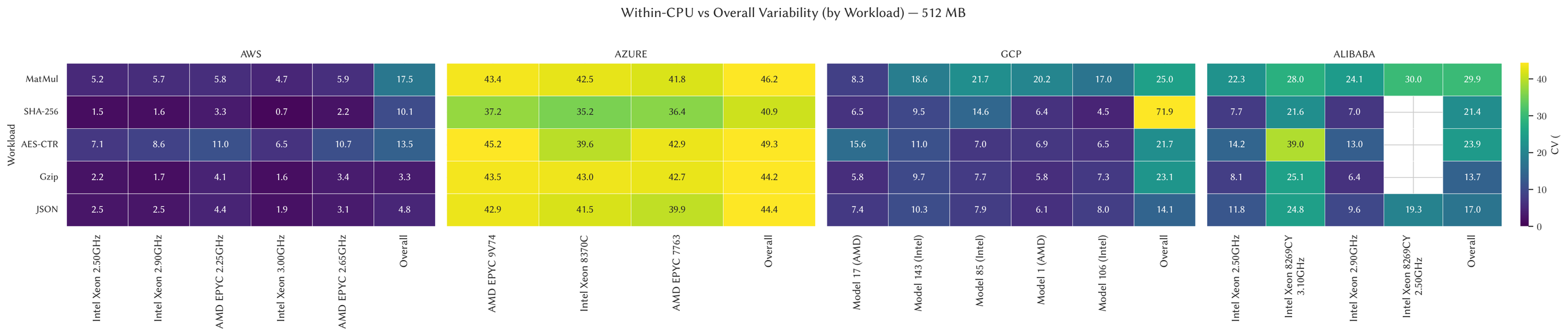

Within-CPU vs Overall Variability (by Workload) — 512 MB
AWS
AZURE
GCP
ALIBABA
Workload
MatMul
SHA-256
AES-CTR
Gzip
JSON
5.2 5.7 5.8 4.7 5.9 17.5
1.5 1.6 3.3 0.7 2.2 10.1
7.1 8.6 11.0 6.5 10.7 13.5
2.2 1.7 4.1 1.6 3.4 3.3
2.5 2.5 4.4 1.9 3.1 4.8
43.4 42.5 41.8 46.2
37.2 35.2 36.4 40.9
45.2 39.6 42.9 49.3
43.5 43.0 42.7 44.2
42.9 41.5 39.9 44.4
8.3 18.6 21.7 20.2 17.0 25.0
6.5 9.5 14.6 6.4 4.5 71.9
15.6 11.0 7.0 6.9 6.5 21.7
5.8 9.7 7.7 5.8 7.3 23.1
7.4 10.3 7.9 6.1 8.0 14.1
22.3 28.0 24.1 30.0 29.9
7.7 21.6 7.0 21.4
14.2 39.0 13.0 23.9
8.1 25.1 6.4 13.7
11.8 24.8 9.6 19.3 17.0
Intel Xeon 2.50GHz
Intel Xeon 2.90GHz
AMD EPYC 2.25GHz
Intel Xeon 3.00GHz
AMD EPYC 2.65GHz
Overall
AMD EPYC 9V74
Intel Xeon 8370C
AMD EPYC 7763
Overall
Model 17 (AMD)
Model 143 (Intel)
Model 85 (Intel)
Model 1 (AMD)
Model 106 (Intel)
Overall
Intel Xeon 2.50GHz
Intel Xeon 8269CY 3.10GHz
Intel Xeon 2.90GHz
Intel Xeon 8269CY 2.50GHz
Overall
CV (
0
10
20
30
40